PID Control of a 1 DOF Helicopter

A Single-Input Single-Output Control System Implementation

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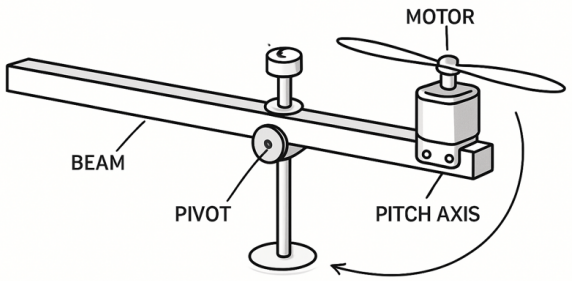
Introduction

Project Overview

- **Goal:** Stabilize a 1 DOF helicopter beam in a horizontal position
- **System Components:**
  - Mechanical beam with pivot (1 degree of freedom - pitch)
  - Motor with propeller providing thrust
  - Potentiometer for angle feedback
  - Arduino microcontroller with LINX firmware
  - LabVIEW control application with PID algorithm

- **Importance:** Understanding control of inherently unstable systems is fundamental to aerospace applications, robotics, and advanced control theory



1 DOF HELICOPTER

System Analysis

Input/Output Variables

- **Input (Manipulated Variable):** Motor control signal (Pulse Width: 1400-2200 µs)
- **Output (Controlled Variable):** Beam angle/position (measured as voltage from potentiometer)
- **Feedback Signal:** Potentiometer voltage (target setpoint: 2.90V)

Control Challenges

- **Inherent Instability:** System naturally falls to one extreme position without active control
- **Non-linearity:** Relationship between motor input and beam angle is not perfectly linear
- **Sensor Noise:** Potentiometer readings affected by motor vibrations
- **Actuator Limits:** Motor speed/thrust has minimum and maximum limits

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Design Approach

Control Strategy: PID Controller

- **Proportional (P):** Provides immediate response proportional to current error
- **Integral (I):** Eliminates steady-state error by accumulating past errors
- **Derivative (D):** Provides damping to reduce overshoot and oscillations

Implementation Choices

- **Feedback Sensor:** Potentiometer (simple, direct voltage output)
- **Platform:** LabVIEW with Arduino (graphical programming, easy visualization)
- **Interface Method:** LINX Toolkit
  - Initially considered VISA serial communication
  - Switched to LINX for simplified hardware interaction
  - Eliminated need for custom Arduino code

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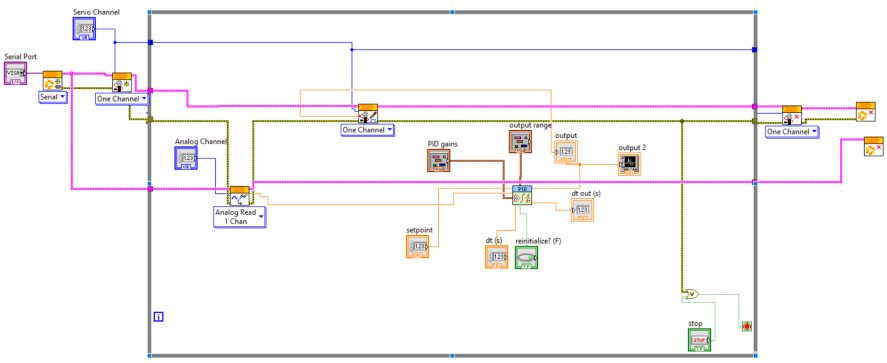
LabVIEW Implementation Overview

Block Diagram Structure



Key Components

- **Initialization:** LINX Open VI (COM7)
- **Feedback:** LINX Analog Read VI (Channel 1/Pin A1)
- **Control:** PID VI with adjustable gains
- **Actuation:** LINX Servo VI (Channel 9)
- **User Interface:** Controls for setpoint (2.90V), PID gains, output range (1400-2200)

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System Operation & Demonstration

Key Observations

- System successfully stabilizes the beam at the horizontal position
- Controller responds to disturbances by adjusting motor thrust
- Beam maintains position with minimal oscillation
- PID output (1897.96 µs shown in front panel) maintains stability

Control Challenges Encountered

Initial Instability

- Early tuning attempts with high gains (P=5.0, Ti=0.001) caused wild oscillations
- System would rapidly swing between extremes without settling

Integral Windup

- Aggressive integral action accumulated large error values
- Caused severe overshoot that persisted even after gain reduction
- System would drive to limits and fail to recover

Sensor Noise

- Motor vibrations affected potentiometer readings
- Created noisy feedback signal that complicated control

Tuning Complexity

- Finding optimal balance between response speed, stability, and accuracy
- Interdependence of P, I, and D terms complicated manual tuning

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Solutions Developed

Systematic Tuning Approach

- **Reset Strategy:** Restart VI to clear PID state (especially integral term)
- **Sequential Tuning:**
  1. Start with low P gain, zero I and D
  2. Incrementally increase P until reasonable response speed
  3. Gradually decrease Ti (increase I action) to reduce steady-state error
  4. Add small D term to dampen oscillations

Visualization Tools

- Added output graph to monitor PID output behavior
- Visual feedback crucial for understanding system response

Final Tuned Parameters

Parameter	Value	Effect
Proportional Gain (P)	3.800	Primary response strength
Integral Time (Ti)	0.038 min	Steady-state error elimination
Derivative Time (Td)	0.002 min	Oscillation damping

Conclusion

Project Achievements

- Successfully implemented PID control of an inherently unstable 1 DOF helicopter system
- Achieved stable horizontal positioning with minimal oscillation
- Developed effective tuning methodology for PID parameters

Key Learnings

- PID tuning is an iterative process requiring systematic approach
- Integral windup can severely impact system performance
- LINX toolkit simplifies hardware interaction in LabVIEW
- Visualization tools are essential for effective tuning

Future Improvements

- Explore alternative sensors (IMU) for improved noise immunity
- Implement digital filtering to reduce sensor noise
- Develop mathematical model for simulation-based tuning
- Explore more advanced control strategies (e.g., state feedback)

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